

VIVEK BHARADWAJ ARYASOMAYAJULA

ENERGY MARKET ANALYTICS | FORECASTING | TRADING ANALYTICS

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PROFESSIONAL SUMMARY

Energy Market Analyst specialising in the Australian National Electricity Market (NEM), with experience across electricity market analytics, forecasting, renewable integration, and operational data analysis. Skilled in **Python, SQL, PostgreSQL, Power BI, and time-series forecasting** using AEMO market data. Experienced in analysing dispatch pricing, market volatility, renewable generation impacts, battery storage analytics, and operational market behaviour. Strong exposure to forecasting workflows, market operations analytics, and renewable energy market analysis using **AEMO datasets and PLEXOS Cloud**.

CORE SKILLS

Energy Markets: AEMO NEM Data, Dispatch Pricing, NEM Fundamentals, FCAS Markets, Demand Analysis, Renewable Integration, Price Volatility, Battery Storage Analytics

Forecasting & Analytics: Time-Series Forecasting, Market Behaviour Analysis, Feature Engineering, Scenario Analysis, Trading Analytics, Operational Analytics

Technical Skills: Python, Pandas, NumPy, XGBoost, TensorFlow/Keras, SQL, PostgreSQL, JavaScript

Visualisation & Modelling: Power BI, DAX, Tableau, PLEXOS Cloud

PROFESSIONAL EXPERIENCE

Research Assistant

Mar 2026 – Present

Transforming Energy Markets (TEM) Research Centre, Sydney

- Maintain and process large-scale AEMO electricity market datasets using Python, SQL, and PostgreSQL for research and operational analytics
- Support forecasting, market operations analytics, and market modelling workflows through data extraction, validation, transformation, and time-series analysis
- Develop Power BI dashboards and analytical reporting tools for electricity market insights, renewable integration analysis, and policy research
- Contribute to analytical workflows involving dispatch pricing behaviour, demand trends, renewable generation, and market volatility analysis

Business Analyst – Finance & Operations

Mar 2025 – Apr 2026

Western Sydney University, Sydney

- Developed Power BI dashboards improving visibility of financial and operational performance metrics
- Analysed workflows to identify process improvement opportunities across finance and administration
- Delivered data-driven insights supporting operational planning and stakeholder decision-making

ICT Business Analyst

Nov 2022 – Jun 2024

CentraHub Pvt Ltd, India

- Automated operational workflows using SQL, reducing response times by 35%
- Developed KPI dashboards supporting business operations across India, Middle East, and Canada regions
- Designed analytics-driven workflows improving operational efficiency and reporting visibility

- Optimised HVAC and energy-intensive utility systems to improve operational efficiency and performance
- Conducted energy performance analysis and monitoring of industrial utility systems
- Worked with SCADA/PLC systems for operational monitoring, validation, and compliance documentation

ENERGY MARKET ANALYTICS & TRADING PROJECTS

NEM Market Intelligence & BESS Trading Platform

- Developing an end-to-end energy market analytics platform integrating AEMO data pipelines, forecasting models, and operational dashboards
- Designing BESS trading strategies to simulate arbitrage opportunities using NEM price signals and operational constraints

Electricity Demand Forecasting (NEM)

- Developed ARIMA, XGBoost, and LSTM models for short-term electricity demand forecasting using AEMO datasets
- Engineered time-series features capturing seasonality, demand peaks, and operational patterns

Electricity Price Forecasting & Market Behaviour

- Built LSTM-based models for 5-minute electricity price forecasting and volatility analysis
- Analysed negative pricing events, renewable intermittency, and supply-demand imbalance impacts

AEMO NEMWeb Data Pipeline

- Built automated ETL pipelines to extract, process, validate, and structure AEMO NEMWeb datasets
- Loaded structured datasets into PostgreSQL environments supporting forecasting and operational analysis
- Developed a web-based portfolio showcase demonstrating the ETL workflow, PostgreSQL integration, and market analytics pipeline architecture

Project Demo: vivekarya05.github.io/nemweb-etl-pipeline/

Selected Technologies: Python, SQL, PostgreSQL, Power BI, XGBoost, LSTM, AEMO NEMWeb, PLEXOS Cloud

CERTIFICATIONS & MARKET TRAINING

- NEM Foundations eLearning Program – **Australian Energy Market Operator (AEMO)**
- Modelling Power Systems Using PLEXOS Cloud – **Energy Exemplar**

EDUCATION

Macquarie University, Sydney

Master of Business Analytics

May 2026

GMR Institute of Technology, India

Bachelor of Technology in Power Engineering

May 2019

Symbiosis College of Distance Learning, India

PG Diploma in Project Management

May 2024